



FEATURE SELECTION OPTIMIZATION WITH META-HEURISTIC ALGORITHMS FOR DISEASE CLASSIFICATION

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Abstract

In this study, there are two different data sets which are the test results of patients for Parkinson disease classification and Cardiac Arrhythmia disease classification. The common problem with those data sets is having excessive number of features. Feature selection is the process of reducing redundant characteristics from a data set and using the best subset of features for machine learning or data mining techniques. By discarding the less-informative features, this helps to reduce the time required to train a learning algorithm as well as the amount of storage required. Meta-heuristic algorithms are used for feature selection in this paper. After selection of the features process, the classification algorithms such as K-Nearest Neighbor, Support Vector Machine and Random Forest implemented for the diagnosing the Parkinson Disease and Cardiac Arrhythmia with a high accuracy.

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LIST OF SYMBOLS

a, b, c	Scalars
$\mathbf{x}, \mathbf{y}, \mathbf{z}$	Vectors
$\mathbf{A}, \mathbf{B}, \mathbf{C}$	Matrices
$\forall x$	Universal quantifier: for all x
$\exists x$	Existential quantifier: there exists x
$\sum_{n=1}^N x_n$	Sum of the x_n : $x_1 + x_2 + \dots + x_N$
$\prod_{n=1}^N x_n$	Product of the x_n : $x_1 \cdot x_2 \cdot \dots \cdot x_N$

LIST OF ABBREVIATIONS

i.e.	Id est (Latin: this means)
e.g.	Exempli gratia (Latin: for example)
BBA	Binary Bat Algorithm
CS	Cuckoo Search Algorithm
EO	Equilibrium Optimizer
GA	Genetic Algorithm
GSA	Gravitational Search Algorithm
GWO	Grey Wolf Optimizer
HS	Harmony Search
MA	Mayfly Algorithm
PSO	Particle Swarm Optimization
RDA	Red Deer Algorithm
SCA	Sine Cosine Algorithm
WOA	Whale Optimization Algorithm
PCC	Pearson Correlation Coefficient
SCC	Spearman Correlation Coefficient

1 Introduction

Computer science technologies such as artificial intelligence, machine learning, advanced algorithms are used in wide fields to improve the life quality of people. One of the fields most affected by technology is medicine. Doctors are diagnosing, treating or improving the quality of the patients' life by using the benefits of the technology. Scientists are trying to support doctors by working more on computer science technologies nowadays. Especially in the COVID-19 pandemic, scientists more focused on the field of medicine. There are many diseases which can be diagnose earlier with computer science. In this article, machine learning and meta-heuristic algorithms are used to diagnose Parkinson and Cardiac Arrhythmia diseases by analyzing the test results of the patients.

Parkinson disease is the most common and the fastest growing neurodegenerative movement disorder, and more than 10 million people live with this disease nowadays and it is increasing rapidly [1]. Statistical results show that Parkinson disease increases with age but 4% of the patients are less than 50 years old [2]. Figure 1 represents the increasing trend of deaths per 100,000 population in United States year by year caused by Parkinson disease.

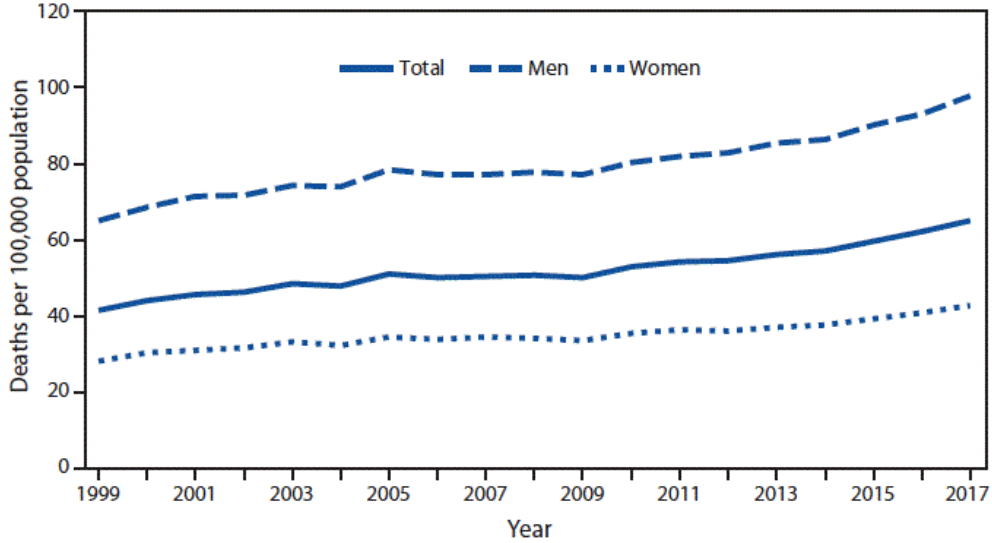


Figure 1: Deaths per 100,000 population in United States caused by Parkinson disease among adults who are greater than 65 years old [3]

Parkinson disease's Tremor and postural symptoms are the most common cardinal motor symptoms, and it also includes other motor and non-motor symptoms like slowness in movement, rigidity, and dysphonia. Moreover,

dopamine is a chemical which is in brain helps to move in a coordination. The neurons which include dopamine called dopaminergic neurons and in the first diagnoses stages, a significant proportion of dopaminergic neurons has already been lost and therefore, early diagnose is vitally important [1]. It is not totally preventable disease in early stages but diagnosing in early stages may improve the life quality of the patients. There are also some symptoms that are correlated with Parkinson disease and reduces the life quality of the patients which are depression, pain, fatigue, and constipation. Parkinson disease has a different progress in every patient. According to this, the treatment should also change patient by patient and drug treatment should start as soon as possible. While diagnosing the Parkinson disease, walking and vocal analysis are the ones mostly used [4].

On the other hand, Cardiac Arrhythmia is a disease caused by the hearth's disrupted normal rhythm. In a daily life, the hearth pumps blood ,which includes nutrients, immune cells, regulatory molecules and oxygen, to the body organs. The rhythm of the hearth is determined by a part of cardiac muscle cells and those cells are controlled by nerves and hormones. The hearth rhythm can be monitored by an electrocardiogram (ECG). This device simply measures the difference of the voltage between some body surface points. According to the abnormal result of the monitoring, doctors are diagnoses patients for different types of cardiac arrhythmia [5].

There are many reasons caused cardiac arrhythmia such as heart attack, block arteries, diabetes, high blood pressure, infection of COVID-19, hyperthyroidism, hypothyroidism, sleep apnea, drinking too much alcohol or caffeine, drug abuse, genetics, smoking, stress or anxiety. Moreover, the symptoms of the cardiac arrhythmia can be fluttering in the chest, racing or slow heartbeat, chest pain or shortness of breath. There are also some indirect and rare symptoms which are anxiety, fatigue and sweating. The classification of the cardiac arrhythmia should be done correctly and then directives should be taken from doctor according to type of it.

In this study, there are two data sets which are the test results of patients for Parkinson disease classification and Cardiac Arrhythmia disease classification. The common problem with those data sets is having too many features. The Parkinson data set includes 754 features and Cardiac Arrhythmia data set includes 279 features. The aim of this paper is that implementing meta-heuristic algorithms for feature selection. Feature selection is the process of reducing redundant characteristics from a data set and using the best subset of features for machine learning or data mining techniques. By discarding the less-informative features, this helps to reduce the time required to train a learning algorithm as well as the amount of storage required. After selection of the features process, the classification algorithms such as K-Nearest

Neighbor, Support Vector Machine and Random Forest implemented for the diagnosing the Parkinson Disease and Cardiac Arrhythmia. Finally, the combination of the meta-heuristic and machine learning algorithms increase the accuracy of the classification.

2 Literature Review

There are many studies used meta-heuristic algorithms as a feature selection method in literature. Enireddy et al. [4] has studied on the same Parkinson disease data set which is used also in this paper. First, it transforms the data set by using Min-Max scaler and then it considers implementing meta-heuristic algorithms to optimize the feature selection. The aim of that paper is using the Gray Wolf Optimization technique as a meta-heuristic algorithm. This technique simply replicates hunting procedure of wolves in wildlife. It creates a social hierarchy between agents of the algorithms like wolf pack and tries to reach optimal solution step by step. There are two different Gray Wolf optimization algorithms used in that article. The main difference between those algorithms is locating the agents according to target and they used the same fitness function which is the number of correctly classified samples divided by the total number of samples. After the feature selection process, there are many classifier machine learning algorithms used such as Extreme Gradient Boosting (XGB) Classifier, k-Nearest Neighbor Classifier, Support Vector Machine Classifier etc. After the computation, it states that the best accuracy is 97.96% with XGB-Classifier.

On the other hand, Guvenir et al. [6] published the Cardiac Arrhythmia data set which is used in this paper. The authors offer a novel machine learning technique for diagnosing cardiac arrhythmia using normal 12-lead ECG records. Voting Feature Intervals (VFI5) is the name of the algorithm. VFI5 is a supervised and inductive learning technique that uses examples to infer categorization expertise. A training set of records is sent into VFIS. Each record includes clinical measures derived from ECG signals as well as other information such as sex, age, and weight, as well as an expert cardiologist's conclusion. The knowledge representation is based on a new approach known as Feature Intervals, in which an idea is represented by the projections of the training examples on each feature individually. In VF15, classification is determined by a majority vote among the class predictions generated by each feature individually. Authors also compared the proposed algorithm with Naive Bayesian and Nearest Neighbor algorithms.

Furthermore, Oreski et al. [7] studied on a hybrid genetic algorithm with neural networks. The developed algorithm is used for optimization of credit

risk classification and real-world credit data set is used. It states that nature inspired algorithms are simply uses neural network classification algorithm as a fitness function and it measures the success of the agents by using accuracy. The results show that feature selection improves the classification accuracy and hybrid genetic algorithm performance better than a simple genetic algorithm.

3 Methodology

3.1 Design

In the design stage of this paper, the data sets are gathered from UCI Machine Learning Repository. After gathering, the data is cleaned and then normalized by using Min-Max Scaler. The meta-heuristic algorithms are implemented to those cleaned data sets and inside the meta-heuristics, machine learning classification algorithms are used for fitness calculation. Figure 2 shows the mentioned design with a diagram.

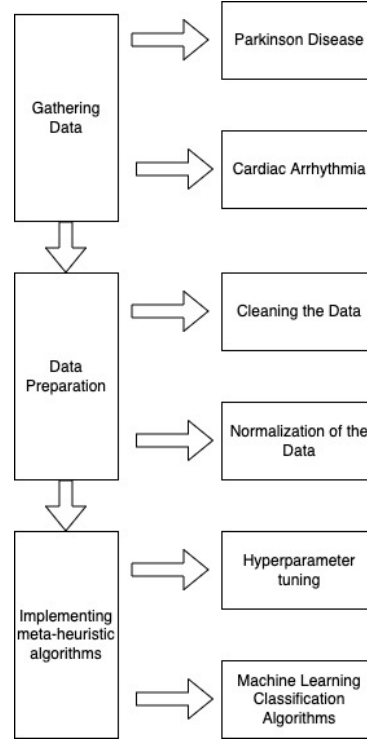


Figure 2: Design Diagram

3.2 Data

The Parkinson data set is collected at Department of Neurology in Cerrahpaşa Faculty of Medicine, Istanbul University. The data set includes 756 instances and 754 attributes. The instances are collected from 188 patients and 64 healthy individuals. Patients have already diagnosed with Parkinson disease and their ages are between 33 to 87. On the other hand, healthy individuals' ages are between 41 to 82. The attributes are related with the numeric data of the voices and the class of the data set is binary which shows the individual has Parkinson disease or not. While collecting the data, microphone is set to 44.1 KHz and each individual was asked to repeat the sustained phonation of the vowel /a/ three times [8].

The Cardiac Arrhythmia data set consists of 452 patients records for 279 features. There are 16 different classes which are 01 refers to normal ECG, 02 to Ischemic changes (Coronary Artery Disease), 03 to Old Anterior Myocardial Infarction, 04 to Old Inferior Myocardial Infarction, 05 to Sinus tachycardia, 06 to Sinus bradycardia, 07 to Ventricular Premature Contraction, 08 to Supraventricular Premature Contraction, 09 to left bundle branch block, 10 to right bundle branch block, 11 to first degree AtrioVentricular block, 12 to second degree AtrioVentricular block, 13 to third degree AtrioVentricular block, 14 to left ventricle hypertrophy, 15 Atrial Fibrillation or Flutter, and 16 refers to the rest. Features include the some personal informations such as age, sex, height, weight and the large portion of the features are the numeric data of the ECG results. The data set includes missing values and class distribution is not fair [6].

3.3 Methods

Meta-heuristic algorithms are problem-independent frameworks that operates on previously specified strategies to achieve optimization. Important examples of meta-heuristics are Genetic Algorithm, Gray Wolf Optimizer, Particle Swarm Optimizer and so on. Meta-heuristic algorithms are mainly impressed by the natural environments. The term "Meta-heuristic" was first introduced by Glover (1986), and it is a combination of two Greek words.

Meta-heuristic algorithms always operate on the basic heuristic approaches, although they are customized for different strategies. In this way, they are distinguished from exact solution approaches, because in exact methods, an optimal solution will be reached, and this usually happens after a long computational period. Therefore, meta-heuristics were developed to obtain a substantially good solution in a relatively short amount of time.

The difference between meta-heuristics and exact methods can be ob-

served best in complicated models with heavy data loads because meta-heuristics will be able to obtain better solutions with less error and this will be done in a substantially shorter time period. Also, meta-heuristics are more flexible than exact methods. They can be customized and fitted to real life scenarios with great ease, which is another superiority over exact methods, because those methods are generally more rigid. In this study, 12 different meta-heuristic algorithms selected to implement both data sets for feature selection such as Binary Bat Algorithm, Cuckoo Search Algorithm, Equilibrium Optimizer, Genetic Algorithm, Gravitational Search Algorithm, Gray Wolf Optimizer, Harmony Search, Mayfly Algorithm, Particle Swarm Optimizer, Red Deer Algorithm, Sine Cosine Algorithm, and Whale Optimization Algorithm.

Binary Bat Algorithm (BBA) is simply inspired by the echolocation behavior of bats. The search procedure is carried out utilizing artificial bats as search agents, matching the natural pulse loudness and emission rate of actual bats, as inspired by the echolocative behavior of bats. The original Bat Algorithm works on continuous real search spaces but Binary Bat Algorithm works on discrete search spaces like feature selection. When chasing prey, bats tend to lower the volume and increase the pace at which ultrasonic sound is emitted. This behavior is mathematically modeled for mimic it in computational environment. There are three main vectors such as position vector, velocity vector, and frequency vector of the agents are updated in the iterations. In the discrete binary space, the position change can be represented by being "0" or "1". This change is directly related with the velocity of the agents. This change represents the feature selection process of the algorithm. The success of the change may evaluate with different fitness functions [9]. Figure 3 represents the general flow chart of Bat Algorithm.

Cuckoo Search Algorithm (CSA) is simply inspired by cuckoo breeding behavior. They have very aggressive reproduction strategy. Some cuckoo species lay their eggs in communal nests, yet they may remove other cuckoos' eggs to maximize the chances of their own eggs hatching. This is called brood parasitism and it is divided into three such as intraspecific brood parasitism, cooperative breeding, and nest takeover. If a host bird realizes the eggs are not its own, it will either throw them away or abandon the nest and construct a new one somewhere else. Three different rules are accepted for the simplicity of the Cuckoo Search Algorithm. First rule is each cuckoo lays one egg at a time and the nest is picked at random. The second rule is that next generations will include the previous generation's best nests with high quality of eggs (solutions). The last rule is that the number of host nests available is set and the owner of the nest can realize the parasite egg with a probability of P_a . After that, the owner of the nest will choose to accept

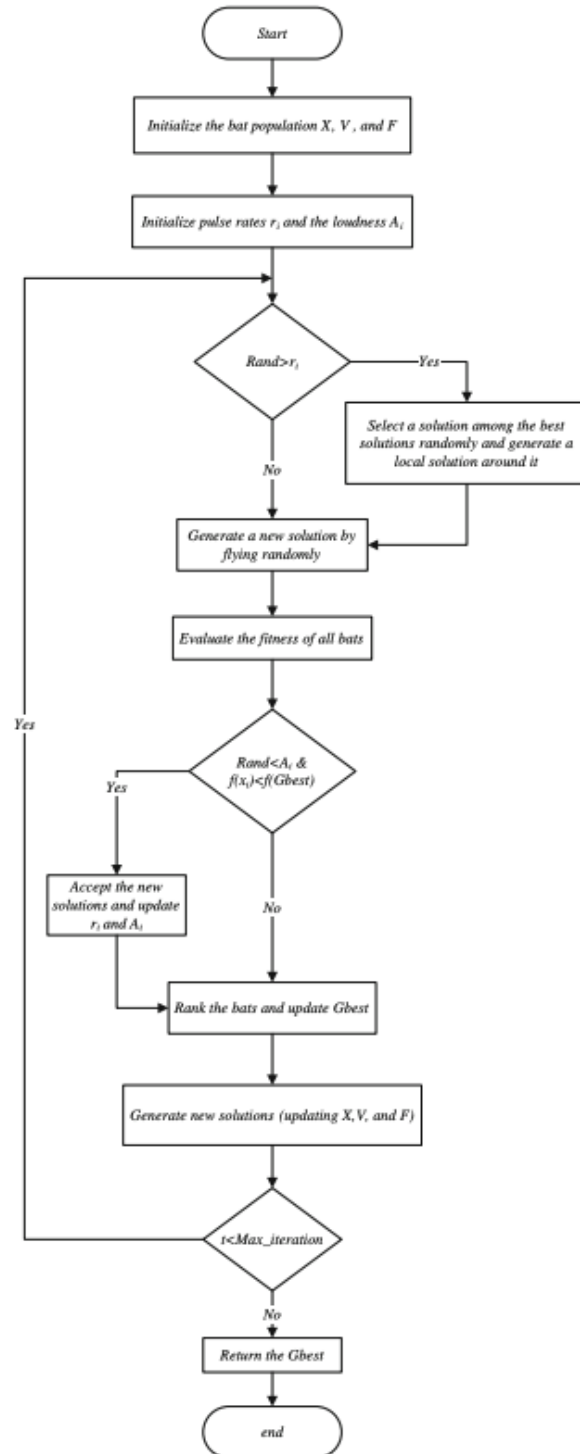


Figure 3: Flow chart of Bat Algorithm[9].

it or reject it. This simply creates diversification in the iterations. The success of the eggs are calculated by fitness function and the new solutions are generated by Lévy flights which is a kind of method that works with Lévy distribution[10]. The pseudo-code of the Cuckoo Search Algorithm is showed in the Algorithm 1.

Algorithm 1 Cuckoo Search Algorithm[10]

- 1: Objective function $f(x), x = (x_1, \dots, x_d)^T$;
 - 2: Initial population of n host nests $x_i = (i = 1, 2, \dots, n)$;
 - 3: **while** ($t < \text{Maximum Generation}$) or (stop criterion); **do**
 - 4: Get a cuckoo (say i) randomly and generate a new solution by Lévy flights;
 - 5: Evaluate its quality/fitness; F_i
 - 6: Choose a nest among $n(\text{say } j)$ randomly;
 - 7: **if** $F_i > F_j$ **then**
 - 8: Replace j by the new solution;
 - 9: **end if**
 - 10: Abandon a fraction (P_a) of worse nests [and build new ones at new locations via Lévy flights];
 - 11: Keep the best solutions (or nests with quality solutions);
 - 12: Rank the solutions and find the current best;
 - 13: **end while**
 - 14: Post process results and visualization;
-

Genetic Algorithm (GA) is the most critical meta-heuristic algorithm in this paper. The reason behind that the best accuracy is gathered from genetic algorithm for both Parkinson and Arrhythmia data sets. Genetic algorithm, which as a sub-class of evolutionary algorithms, is a perfect example of combined interdisciplinary studies insomuch as being a heuristic that imitates the biological process of evolution itself. Referencing the natural evolution theory of Charles Darwin, who fundamentally describes it as “descent with modification”, GA copies the theory to be applied in computer science and OR. The “survival of the fittest” process, where the fittest offspring gets to survive and pass on its genotype to the future generations, is applied to GA. Parents who have higher fitness score have also higher chance to create off-springs. Therefore, the new populations will have totally higher fitness scores. An initial population, which is a set of random initial solutions in GA’s case, is required to start the iterations. This initial solution also may be a result of another algorithm. Each individual in the algorithm is defined by a set of variables, denoted as strings, usually as binary values, which are

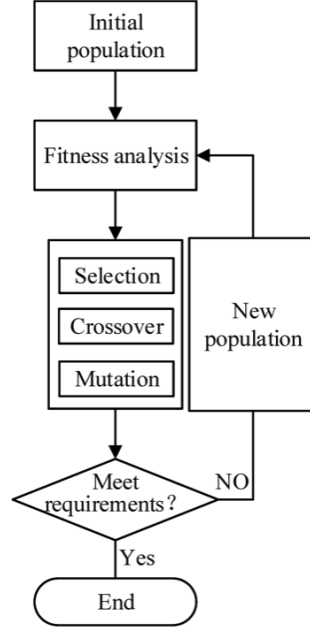


Figure 4: Flowchart of Genetic Algorithm [11]

called genes. Genes come together to create chromosomes, which is actually the solution coded in strings. After selecting the initial population, those parents are used for the cross-over phase by selecting one (or more than one) random cross-over point within the genes to swap the genetic codes in order to create off-springs. Throughout this whole process, each individual is assigned a fitness score by using the fitness function to the extent of obtaining computational results that gives an understating. The main aim is that finding the best fitted agent in the populations. If the case is the selection of the features, the fitness function may be the machine learning classification algorithms and the score of the agent can be measured by the accuracy of the classification. Figure 4 represents the general flow chart of Bat Algorithm.

Gravitational Search Algorithm (GSA) is inspired by the law of gravity and mass interactions. In the general form of GSA, the performance of the agents are measured by their masses. The gravitational pull attracts all of these agents, causing an universal movement of all agents towards the agents with heavier masses. The heavy masses (good solutions) move more slowly than lighter agents. This rule simply provides the success of the next steps. There are four main important attributes of the agents which are position, inertial mass, active gravitational mass, and passive gravitational mass. The gravitational and inertial masses are calculated with fitness function of the algorithm and they are directly affect the position algorithm. The position

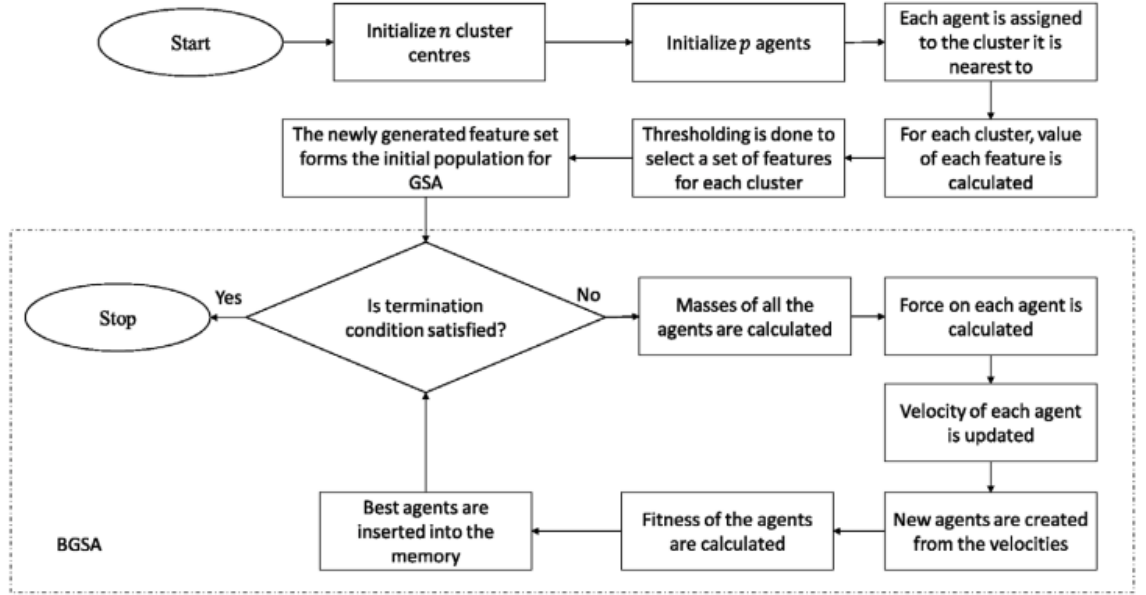


Figure 5: Flowchart of GSA[13]

represents the solution of the problem. The expectation when the number of iteration increases is that masses be attracted by the heaviest mass [12]. Figure 5 represents the general flow chart of Bat Algorithm.

Grey Wolf Optimizer (GWO) is inspired by the hunting process of the grey wolf packs. There are four hierarchical category of the wolfs in a pack. First one is alpha, the alpha wolf is the leader of the pack and gives the decisions. The second level is betas, they simply helps alpha wolf to give decision or other daily pack activities. When the alpha wolf gets older or passes away, one of the beta wolf gets alpha. Third level is omega wolves, they are required to obey to all other dominant wolves at all times. The last one is omega wolves which includes scouts, sentinels, elders, hunters, and caretakers. They are more dominant than the omega wolves [14]. Those hierarchical categories represent the success of the solution in the mathematical model. The best solution is alpha, than beta and delta follow respectively. All of the other solutions are omega. According to the position (solution) of the alpha, beta or delta wolves affect the position of the delta wolves. The success of the position is directly related with the fitness function. The fitness function is selected from classification algorithms such as Support Vector Machine or Random Forest.

Harmony Search (HS) is inspired by the harmony of the music. It is directly related with the idealized improvisation process by a skilled musician.

There are three possibilities when improvising. First one is playing exactly the same thing which is memorized before, second one is playing a similar thing which is memorized before and the last one is playing random notes. Zong Woo Gem et al. formalized those three options into usage of harmony memory, pitch adjusting, and randomization [15]. Usage of harmony memory is very similar with choosing the best-fitted agent in GA to reproduce it for next generations. Acceptance of this harmony is directly related with a probability parameter r_{accept} . If this rate is too high, the algorithm can not diversified because it stuck with previously defined harmonies. Pitch adjustment means changing the solution slightly. This process is very similar with mutation operation in GA. This helps the diversification of the solution space. r_{pa} represents the pitch-adjusting rate in HS Algorithm. This rate affects directly the intensification and diversification of the problem. The third one is randomization which is again related with intensification and diversification but the difference between pitch-adjusting and randomization is that randomization diverse the solution more [16]. The pseudo-code of the HS Algorithm is showed in the Algorithm 2.

Algorithm 2 Harmony Search Algorithm [16]

- 1: Objective function $f(x), x = (x_1, \dots, x_d)^T$;
 - 2: Generate initial harmonics (real number arrays)
 - 3: Define pitch adjusting rate (r_{pa}), pitch limits and bandwidth
 - 4: Define harmony memory accepting rate (r_{accept})
 - 5: **while** ($t < \text{Max number of iterations}$) **do**
 - 6: Generate new harmonics by accepting best harmonics
 - 7: Adjust pitch to get new harmonics (solutions)
 - 8: **if** ($rand > r_{accept}$) **then**
 - 9: choose an existing harmonic randomly
 - 10: **else if** ($rand > r_{pa}$) **then**
 - 11: adjust the pitch randomly within limits
 - 12: **else** ($rand > r_{pa}$)
 - 13: generate new harmonics via randomization
 - 14: **end if**
 - 15: Accept the new harmonics (solutions) if better
 - 16: **end while**
 - 17: Find the current best solutions
-

Mayfly Algorithm (MA) is inspired by the mayflies' social behaviors and mating process. Position of the mayflies shows the possible +solution in the solution space. First, two sets which are male and female mayflies are generated in this algorithm. After the creation, they are randomly located in

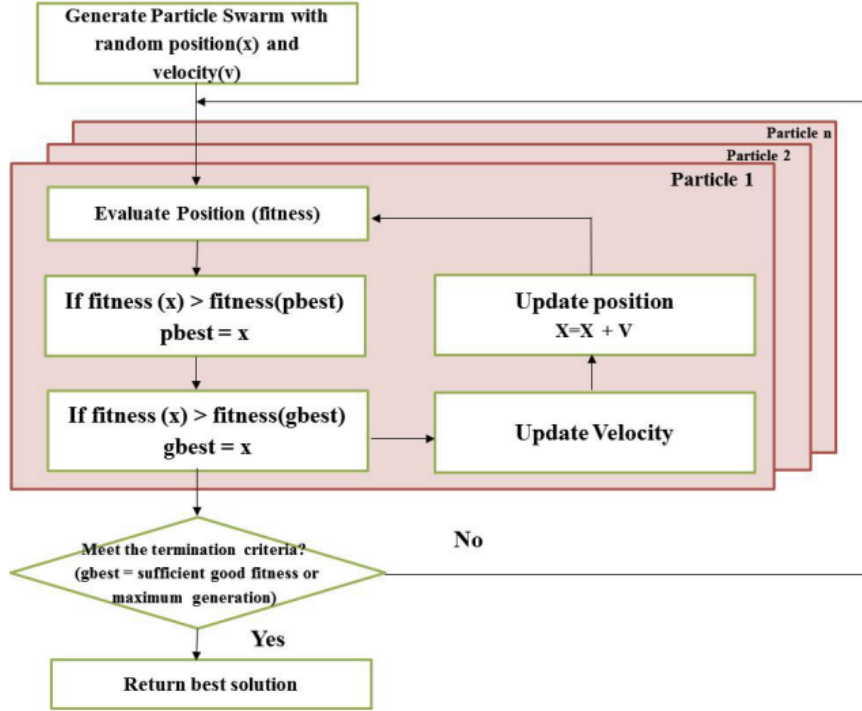


Figure 6: PSO Process[18]

the solution space and the performance of them may calculated with the objective function. Velocity and the flying direction are two parameters which directly affect the solution. Each mayfly arranges its trajectory towards the best mayfly [17].

Particle Swarm Optimization (PSO) is inspired by the behavior of the flying birds and their relationship to solve the problems. The PSO can check the big solution spaces because every particle may explore different spaces. PSO has advantage to use for feature selection because the algorithm has memory and can work on binary or discrete data. The main parameters for this algorithm is position (x) and velocity (v) [18]. Figure 6 represents the process of PSO.

Red Deer Algorithm (RDA) is inspired by the breeding season process of the red deer. The RDA algorithm also starts with a random initial population like GA. Best Red Deers (RD) are selected according to fitness function and named as male RD and other ones named hinds. The male RD should roar first and foremost. They are separated into two classes based on how powerful a roaring phase is. Those classes are commanders and stags. After then, each harem's commanders and stags fight it out to own their harem. Harems are

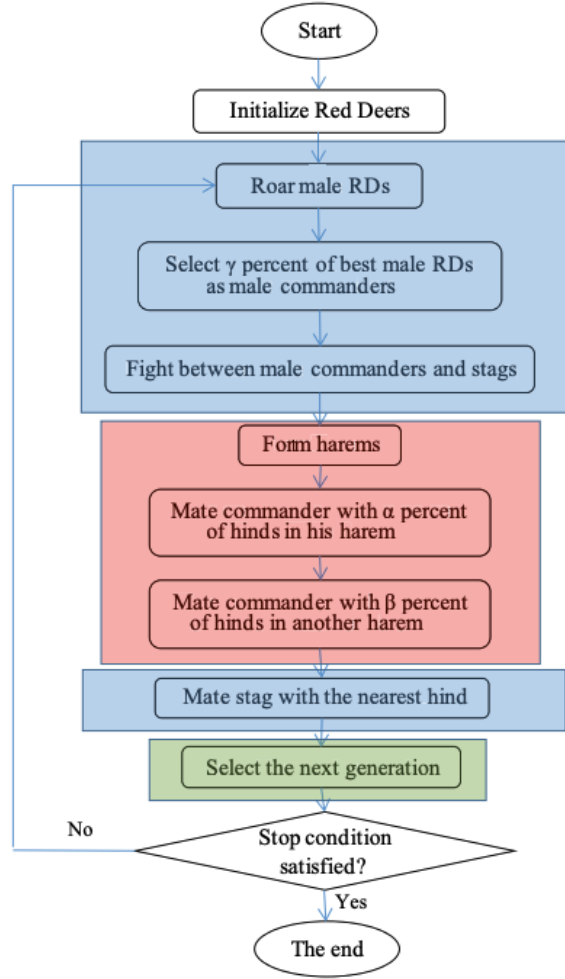


Figure 7: Red Deer Algorithm Flowchart [19]

also founded by commanders. The quantity of hinds in harems is proportional to the commanders' roaring and fighting ability. As a result, commanders in harems mate with a large number of hinds. The other males mate such as stags with the nearest hind without regard for the harem's limitations. The user can arrange the phases in terms of the parameters used and the mathematical formulation. Figure 7 shows the flowchart of the RDA.

Whale Optimization Algorithm (WOA) is inspired by the hunting process of the humpback whales. Humpback whales hunt krill or small fish in groups, encircling and trapping their prey in bubble nets. By considering each whale as a feature subset, foraging behavior is employed for future selection. Binary strings shows the whales in the algorithm and length of the string must

be equal to the total number of features in the data set. In the binary string zero shows the feature is not included and one shows the feature is included. Each whale moves in response to its prey's location or in search of prey. The whales in the used algorithm for this paper attempt to strike a balance between exploration and exploitation in their migrations. This balance is crucial in whales to prevent hazards such as getting stuck in a local optimum or failing to thoroughly explore the search space. Exploitation is the process of identifying a superior answer from a previously studied search area. Exploration is the process of advancing toward previously undiscovered areas of the search space [20]. Finally, Equilibrium Optimizer (EO) and Sine Cosine Algorithms (SCA) are inspired from physical and mathematical rules. EO is simply inspired by the mass balance equation and on the other hand, SCA is simply inspired by the sine cosine functions. They both mimic the equations for changing the solution randomly through high exploration and exploitation [21] [22].

On the other hand, filter-based feature selection methods do not use any intermediate learning algorithm. Instead, they employ statistical methods to determine the relative relevance of features in the context. The features are ranked by the filter-based methods and then top features can be used for classification algorithms. In this study, three different feature-based methods such as Pearson Correlation Coefficient, Spearman Correlation Coefficient and Relief used to show the better performance of the meta-heuristic algorithms.

Pearson Correlation Coefficient (PCC) is a linear correlation measurement between two data. The ratio shows the covariance of two variables and the product of their standard deviations. The ratio is between -1 and 1. If the ratio is close to 1, it means that the relation is strong. Spearman Correlation Coefficient (SCC) describes the relationship of the features with monotonic function. The main difference between PCC and SCC is that PCC looks for linear relationship and SCC looks for monotonic relationship. Finally, Relief is a statistical based feature selection algorithm. Relief assigns each feature a feature score, which can subsequently be used to rank and select the highest-scoring features for feature selection. These scores could also be used as feature weights to aid downstream modeling. The discovery of feature value discrepancies between nearest neighbor instance pairs is used to score relief features. The feature score lowers if a feature value difference is discovered in a surrounding instance pair with the same class. The feature score increases if a feature value difference is found in a surrounding instance pair with different class values.

After implementing the meta-heuristic algorithms or filter-based feature selection methods, machine learning classification algorithms are implemented

such as Support Vector Machine (SVM), K-Nearest Neighbor (K-NN) and Random Forest (RF). Support Vector Machine can solve linear or non-linear problems and the idea behind is simply dividing data into classes with a hyper-plane or a line. It simply draws support vectors between the data points and the line or the hyper-plane. The distance is called margin and the aim of SVM is to maximize the margin. K-Nearest Neighbor is a supervised machine learning algorithm which works again with the distance principle. It is randomly divides the data points into K classes and then according to mid point of the classes, it changes the classes of the data points. The aim is that minimizing the distances between mid point of the classes and data points. Finally, Random Forest algorithm uses large number of decision trees together. Each tree in the random forest produces a class prediction, and the class with the most votes becomes the prediction of our model.

In this paper, a Python package which is developed by Ritam Guha et al. used for implementing 12 different meta-heuristic algorithms and 3 different filter-based methods which described above [23]. Furthermore, a Genetic Algorithm is also developed for this paper. In the developed Genetic Algorithm, agents' genotype includes features as binary strings and a random number for mutation operation. The random number shows the number of features are changed in the mutation operation. The length of the string is equal to the total number of features in the data set. In the binary string zero shows the feature is not included and one shows the feature is included. In the fitness, the algorithm creates a new data set according to 1 values of the agent's genotype. For the score calculation, it implements the classification algorithm and equalize the accuracy of the test set to the score of the agent. For the next generations, the algorithm selects two agents as parents. The selection operation is called roulette wheel selection. It means that selection of highly scored agents is more possible. With those parents, the algorithm creates two child agents with implementing cross-over and mutation. There are two different cross-over function in the algorithm. The first one takes the even numbered genes from the first parent and odd numbered genes from the second parent. The second cross-over function does exact the reverse of the first cross-over function. After getting two child agent from those two functions, it implements mutation according to a probability rate. The mutation function changes ones to zeros or zeros to ones in the genotype for randomly selected features. The total number of changed features is related with the gene of the agent as mentioned before. Finally, the algorithm returns the best agent in the populations.

3.4 Experiment

In the experiment of this study, Python programming language is used in two different development environments which are Google Co-lab and Spyder. The Parkinson data set is clear and therefore, Min-Max Scaler is only implemented for normalization of the data. The Arrhythmia data set has missing values, and therefore column "J" is deleted from the data set and rest of the missin values are filled with "interpolate" function. After that, Min-Max Scaler is implemented for the normalization of the data. The data is also splitted into test and train data sets. The train data set includes the 70% of the total data set and seperation is done with a constant random seed value which is 10.

After the data preparation, meta-heuristic algorithms are implemented. The total population number is set as 100 and number of agents in the population is set as 20. Specifically, the mutation rate for Genetic Algorithm is 20% and the cross-over rate is 100%. After the feature selection, classification algorithms are implemented. The classification algorithms come from the Sklearn Libarary. The accuracy of the algorithms are showed in the Table 1 and 2.

	SVM	KNN	RF
w/o Feature Selection	%85.90	%88	%86.78
Genetic Algorithm	%91.63	%91.63	%91.63
Genetic Algorithm w/Grid Search	%91.63	%98.24	%92.03
Binary Bat Algorithm	%85.46	%87.22	%%87.67
Cuckoo Search Algorithm	%74.89	%80.62	%87.22
Equilibrium Optimizer	%85.46	%87.22	%90.31
Genetic Algorithm	%84.21	%87.5	%88.16
Gravitational Search Algorithm	%86.34	%84.14	%88.11
Gray Wolf Optimizer	%86.34	%83.70	%90.75
Harmony Search	%85.46	%89.87	%89.87
Mayfly Algorithm	%86.78	%89.87	%90.31
Particle Swarm Optimizer	%85.02	%87.67	%88.55
Red Deer Algorithm	%80.62	%85.46	%88.11
Sine Cosine Algorithm	%85.02	%88.55	%91.19
Whale Optimization Algorithm	%86.78	%85.02	%90.31
Pearson Correlation Coefficient	%88.55	%84.58	%91.19
Spearmen Correlation Coefficient	%84.14	%87.67	%86.34
Relief	%85.90	%86.78	%90.75

Table 1: The Parkinson disease accuracies

	SVM	KNN	RF
w/o Feature Selection	%61.03	%58.82	%72.06
Genetic Algorithm	%72.79	%72.79	%80.88
Genetic Algorithm w/Grid Search	%77.94	%69.12	%78.68
Binary Bat Algorithm	%59.56	%57.35	%71.32
Cuckoo Search Algorithm	%60.29	%59.56	%73.53
Equilibrium Optimizer	%61.76	%60.29	%69.12
Genetic Algorithm	%58.08	%58.09	%69.85
Gravitational Search Algorithm	%58.82	%56.62	%68.38
Gray Wolf Optimizer	%58.82	%57.35	%69.12
Harmony Search	%60.29	%60.29	%73.53
Mayfly Algorithm	%58.08	%58.82	%72.79
Particle Swarm Optimizer	%61.03	%59.56	%73.53
Red Deer Algorithm	%62.50	%58.09	%65.44
Sine Cosine Algorithm	%59.56	%58.82	%61.76
Whale Optimization Algorithm	%58.82	%59.56	%65.44
Pearson Correlation Coefficient	%57.35	%55.88	%68.38
Spearman Correlation Coefficient	%62.50	%59.56	%73.53
Relief	%59.56	%61.03	%69.12

Table 2: The Arrhythmia disease accuracies

According to the tables above, the best accuracies both come from the developed Genetic Algorithm for this paper. The best accuracy for the Parkinson disease classification is 98.24% and as mentioned before, Enireddy et al. [4] states that the best accuracy is 97.96% with Gray Wolf Optimizer and XGB-Classifier. On the other hand, the best accuracy in this study for the Cardiac Arrhythmia disease classification is 80.88% and Guvenir et al. [6] states that authors reached 62% accuracy with the proposed algorithm. The reasons behind the increased accuracy are changing the scoring method in the fitness function, the way that creating the new populations, the type of mutation and cross-over. Even the small number of increase in the accuracy may be vital for the patients.

4 Conclusion

The health topic is very popular nowadays due to pandemic. The computational studies focus on to improve the life quality of people. To conclude this paper, there are two different diseases are considered in this paper which are the Parkinson disease and Cardiac Arrhythmia. The reason for implementing

13 different meta-heuristic algorithms is elimination of redundant features in the data sets. The accuracies are achieved with classification algorithms such as Support Vector Machine, K-Nearest Neighbor and Random Forest. The results show that the proposed Genetic Algorithm for this paper gave the best accuracies in both data sets. The best accuracy for Parkinson Disease is 98.24% and 80.88% for Cardiac Arrhythmia.

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